DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

1. TITLE	2. IDENTIFICATION NUMBER
Hull Mounted Sonar Self-Noise Report	DI-HFAC-81203

3. DESCRIPTION/PURPOSE

3.1 The Hull Mounted Sonar Self-Noise Control Report provides consolidated design information used by the Government to verify that Contractor efforts are consistent with the specified Sonar Self-Noise Control Report.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
910503	N SEA 55N3		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A6128

10. PREPARATION INSTRUCTIONS

10.1 Format. The Hull Mounted Sonar Self-Noise Control Report shall be in the Contractor's format.
10.2 Content. The Hull Mounted Sonar Self-Noise Control Report shall identify and characterize machinery and equipment (such as size, location relative to sonar transducers, horsepower, RPM, and capacity) representing potential sources of sonar self-noise (noise-critical machinery). Noise critical machinery shall include propulsion diesel generators, ship service diesel generators, air conditioning machinery, refrigeration plants, air compressors, purifiers, sea connected pumps, non-seaconnected pumps greater than 1/2 hp, sewage treatment machinery complex, engine room exhaust fans, plus other auxiliary machinery units located within 40 feet of sonar transducers and driven by motors of power rating greater than 1/2 hp.
10.2.1 The mounting system shall be characterized for each resiliently mounted noise source. The mount manufacturer, size, model, and loading, and all flexible connections to the mounted unit shall be identified.
10.2.2 Calculations shall be identified which demonstrate that machinery noise isolation and machinery-to-sonar separation distances are such that the total noise level at the sonar requirement in the specifications are met.
10.2.3 Line diagrams and sketches of each system containing noise critical machinery shall be included showing all noise control features to be installed. The diagrams shall contain sufficient installation information on noise control features for training, production accomplishment, and inspection verification.
10.2.4 Results of resonance avoidance analyses for each foundation, as required by the specifications, shall be included. Assumptions and analysis procedures shall be clearly defined and a sample calculation included. Forcing frequencies and calculated foundation resonances shall be identified.
10.2.5 For piping penetrations within 20 feet of sonar transducers or piping penetrations of structural bulkheads below the waterline and adjacent to the sonar receivers, the separation distances from the sonar transducers and distance to the hull shall be identified and the location, type, and extent of any damping treatments shall be described.

11. DISTRIBUTION STATEMENT

Distribution Statement A: Approved for public release; distribution is unlimited.

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